

White Wine Quality: A Logistic Regression Model

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Introduction

In this project, we use a generalized logistic linear regression model to analyze white wine quality using physicochemical measurements from the UCI Wine Quality dataset. Given that logistic regression is designed for binary outcomes, we define a response variable called good, where good = 1 if a wine's quality score is at least 7 and good = 0 otherwise. The white wine dataset contains 4,898 observations and includes 11 physicochemical predictors along with the original quality score. The purpose of this study is to understand which chemical characteristics are associated with a higher probability that a white wine will be classified as good.

We completed the analysis in R using the packages tidyverse, ggpubr, broom, car, MASS, pROC, pscl, and ResourceSelection. After cleaning the data, we split the sample into a 70/30 train/test set, where the training set consisted of 3,428 wines and the test set of 1,470 wines. We then removed 208 influential observations from the training data using Cook's distance, compared a full logistic model with stepwise AIC and BIC models, and selected the BIC model as the final model because it was more parsimonious while maintaining a competitive fit. The final model achieved a training ROC AUC of 0.9018 and a test ROC AUC of 0.7872, and on the test set it produced accuracy 0.742, precision 0.438, recall 0.721, specificity 0.748, F1 score 0.545, and log loss 0.470. For a wine with average values of the predictors in the final model, the estimated probability of being classified as good was 0.0575, with a 95% interval from 0.0474 to 0.0697.

Data description

The dataset used in this analysis is the white wine dataset from the UCI Machine Learning Repository. It contains physicochemical measurements and expert quality ratings for Portuguese "Vinho Verde" white wines. The original data includes 12 variables: fixed acidity, volatile acidity, citric acid, residual sugar, chlorides, free sulfur dioxide, total sulfur dioxide, density, pH, sulphates, alcohol, and quality. For this project, we created an additional binary response variable called good, defined as 1 when quality is at least 7 and 0 otherwise. This re-coding converts the problem into a classification setting suitable for logistic regression.

Further, we detail our procedure for identifying and removing influential observations, using Cook's distance, in the Analysis section.

After checking the data with `sum(is.na())` and `colSums(is.na())`, we found no missing values, so no observations were removed. In the original quality distribution, the counts by score are 20 wines rated 3, 163 rated 4, 1,457 rated 5, 2,198 rated 6, 880 rated 7, 175 rated 8, and 5 rated 9. After re-coding, about 78.36% of wines fall into the good = 0 category and 21.64% fall into the good = 1 category, so the response is somewhat imbalanced but still usable for logistic regression. Since good = 1 is the smaller group (about 1 in 5 wines), we later report ROC/AUC and sensitivity/specificity in addition to overall accuracy so that performance is not judged only by the majority class.

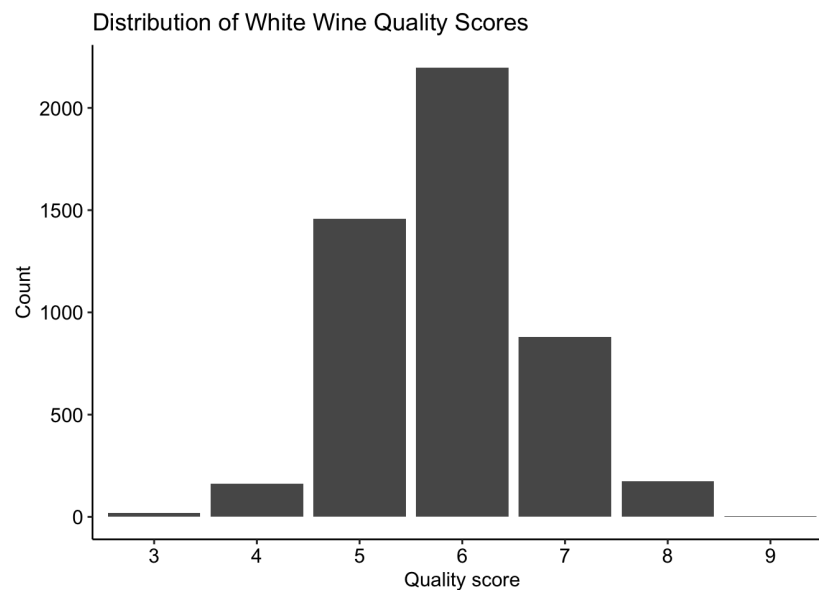


Figure 1: Bar chart of original quality scores (3–9). Most observations are in the middle of the scale, especially quality = 6

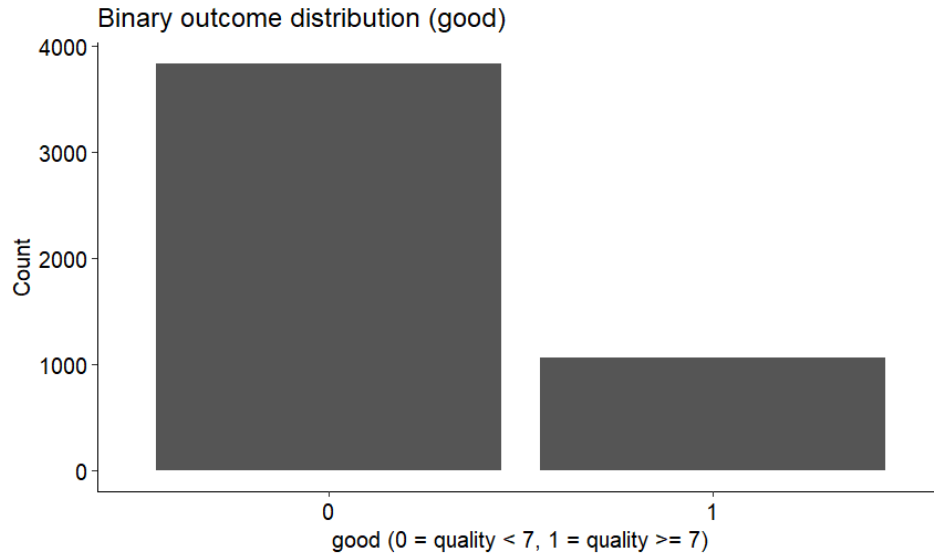


Figure 2: Bar chart of the binary response good (0 = quality < 7, 1 = quality >= 7). The classes are imbalanced, with far more 0's than 1's.

Variable	Description	min/max/mean	Units
<i>Fixed Acidity</i>	Non-volatile acids (mainly tartaric)	3.8 / 14.2 / 6.855	<i>g/L</i>
<i>Volatile Acidity</i>	Volatile acids (mainly acetic)	0.08 / 1.10 / 0.278	<i>g/L</i>
<i>Citric Acidity</i>	Citric acid content	0.0 / 1.66 / 0.334	<i>g/L</i>
<i>Residual Sugar</i>	Sugar remaining after fermentation	0.60 / 65.8 / 6.391	<i>g/L</i>
<i>Chlorides</i>	Salt content	0.01 / 0.35 / 0.046	<i>g/L</i>
<i>Free sulfur dioxide</i>	Free SO ₂ (antimicrobial/antioxidant)	2.00 / 289.0 / 35.31	<i>ppm</i>
<i>Total sulfur dioxide</i>	Total SO ₂ (free + bound)	9.00 / 440.0 / 138.4	<i>ppm</i>
<i>Density</i>	Wine density	0.9871 / 1.039 / 0.994	<i>g/mL</i>
<i>pH</i>	Acidity/alkalinity measure	2.72 / 3.82 / 3.188	<i>pH scale</i>
<i>Sulphates</i>	Sulphate content	0.22 / 1.08 / 0.49	<i>g/L</i>

	(contributes to SO ₂)		
<i>Alcohol</i>	Alcohol by volume	8.0 / 14.2 / 10.51	%
<i>Quality</i>	Expert quality score	3.0 / 9.0 / 5.878	score (0–10)

Analysis

We first cleaned the data by removing missing observations, but because there were no missing values, the dataset remained at 4,898 observations. We then randomly splitted the clean data into a 70/30 training-test partition using a fixed seed. This produced 3,428 observations in the training set and 1,470 in the test set. The class proportions were very similar across the split: in the training set, about 78.27% of wines were in class 0 and 21.73% were in class 1, while in the test set about 78.57% were in class 0 and 21.43% were in class 1. This indicates that the split preserved the overall class balance reasonably well.

Using the training data, we first fit a full logistic regression model with all 11 physicochemical predictors. In this initial model, several predictors were already statistically significant, including fixed acidity, volatile acidity, residual sugar, chlorides, free sulfur dioxide, density, pH, and sulphates, while citric acid, total sulfur dioxide, and alcohol were not significant at conventional levels in the uncleaned full model. Because influential points can distort model estimation, we then computed Cook's distance for the full training model using the common cutoff rule ($4/n$). With ($n = 3428$), the cutoff was approximately 0.00117, and 208 influential observations were flagged and removed from the training set before refitting the model. This left a cleaned training set of 3,221 observations.

After removing influential observations, we refitted the logistic model and then examined diagnostics. We used a residuals-versus-fitted plot to look for major structure in the deviance residuals and a Q-Q plot of deviance residuals as the required diagonal plot. In a logistic model, these diagnostics are only approximate, but they still help assess whether there are severe deviations from model assumptions. The diagnostic plots suggested that the model captures useful structure, although the residual behavior is not perfectly ideal, which is common in classification settings.

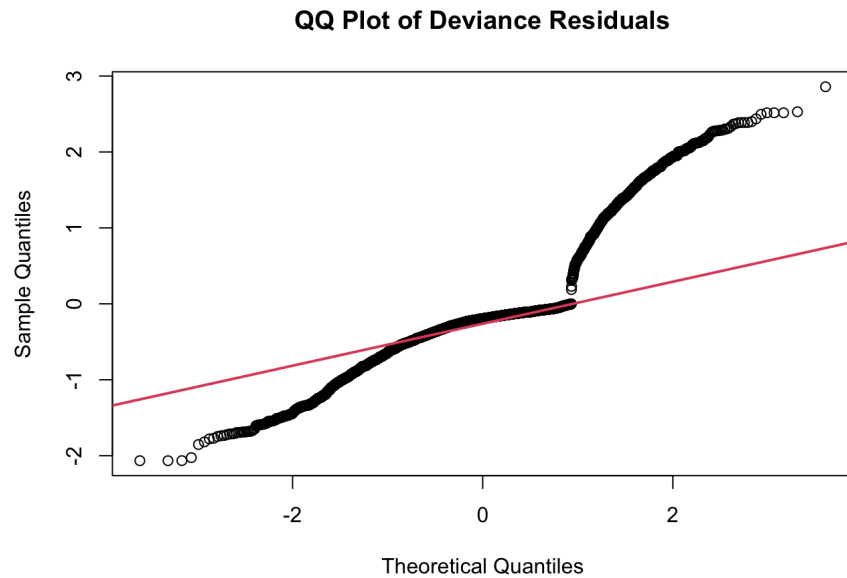


Figure 3: Q–Q plot comparing deviance residuals to theoretical normal quantiles. Deviations from the diagonal line suggest residuals are not well-approximated by a normal distribution

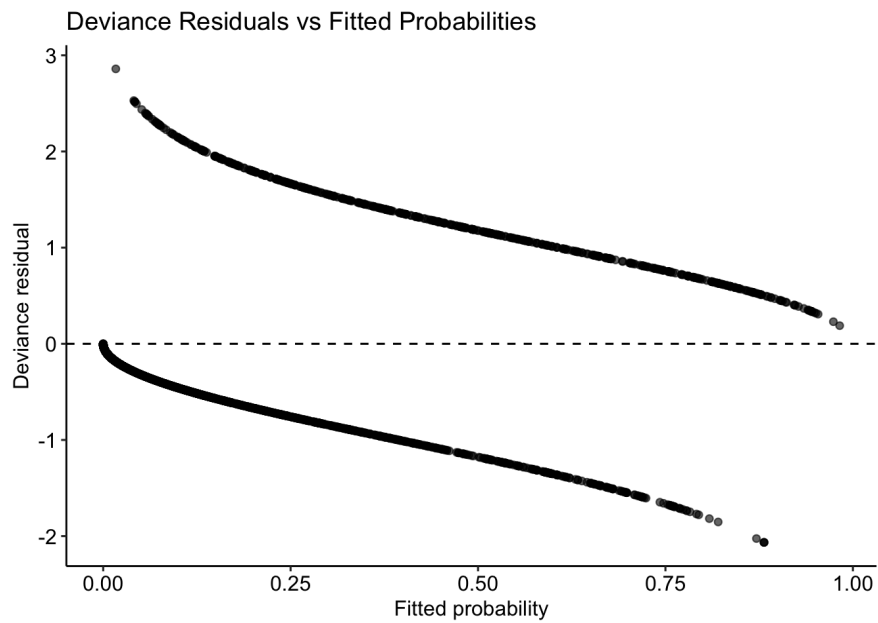


Figure 4: Deviance residuals plotted against fitted probabilities from the final logistic regression model. Two main bands appear due to the binary outcome, with larger residuals indicating misclassified or poorly fit observations.

We also performed a Hosmer–Lemeshow goodness-of-fit test on the cleaned training data and obtained ($X^2 = 37.915$) with 8 degrees of freedom and p-value (7.806×10^{-6}), suggesting some lack of perfect calibration even though the model still has strong discriminatory ability.

Model evaluation and prediction

To identify the best model, we compared three candidate logistic models built on the cleaned training set: the full model with all predictors, a stepwise AIC model, and a stepwise BIC model. The AIC model retained nine predictors, while the BIC model retained eight predictors. The AIC model included alcohol in addition to the BIC set, whereas the BIC model excluded alcohol and retained chlorides, pH, volatile acidity, sulphates, free sulfur dioxide, residual sugar, density, and fixed acidity.

Model	# predictors	AIC	BIC	Deviance	McFadden R ²	Nagelkerke R ²
Full (clean)	11	1865	1938	1841	0.385	0.496
Stepwise AIC	9	1863	1923	1843	0.384	0.496
Stepwise BIC	8	1865	1920	1847	0.382	0.494

Table 2. Model comparison for subset selection (full vs stepwise AIC vs stepwise BIC).

On the cleaned training data, the three models were very close in overall performance. For example, the full model and reduced models had nearly the same training AUC (about 0.902). On the test set, performance was also very similar (test AUC around 0.786–0.787), which indicates that the reduced models preserve most of the predictive value while using fewer variables.

We also used likelihood-ratio tests to compare the reduced models against the full model. For the BIC model versus the full model, the deviance difference was 6.5122 on 3 degrees of freedom with p-value 0.08918. For the AIC model versus the full model, the deviance difference was 1.7876 on 2 degrees of freedom with p-value 0.4091. These results indicate that the reduced models are not dramatically worse than the full model, and they support choosing the BIC model as the final model because it achieves a simpler representation with only a small sacrifice in fit.

The final model was therefore the BIC-selected logistic regression:

good ~ chlorides + pH + volatile_acidity + sulphates + free_sulfur_dioxide + residual_sugar + density + fixed_acidity.

All eight retained predictors were statistically significant in the final fitted model. The fitted coefficients showed that higher pH, sulphates, free sulfur dioxide, residual sugar, and fixed acidity were associated with higher log-odds of being a good wine, while higher chlorides, volatile acidity, and density were associated with lower log-odds of being a good wine, holding the other variables fixed. The final model had null deviance 2991.0, residual deviance 1847.3, and AIC 1865.3 on the cleaned training data.

Using odds ratios makes the coefficients easier to interpret. The estimated odds ratio for pH was about 642, for sulphates about 61.8, for residual sugar about 1.66, for fixed acidity about 2.57, and for free sulfur dioxide about 1.02. The odds ratio for volatile acidity was about 0.0176 and for chlorides was extremely small, indicating that increases in these variables are associated with sharply lower odds of being classified as a good wine. Density also had an odds ratio essentially equal to 0 on this scale. Because some predictors such as pH and density have units where a one-unit change is unrealistically large, the signs and relative directions are more meaningful than the raw one-unit odds-ratio magnitudes. Multicollinearity was not severe for most predictors, although residual sugar and density had the largest VIF values at about 5.16 and 6.30, respectively.

For classification, we selected the threshold using the training ROC curve and Youden's method. The resulting threshold was 0.1484781. The training ROC AUC was 0.9018, indicating strong discrimination on the cleaned training data. On the test set, the BIC model slightly outperformed the full and AIC models. The final BIC model achieved accuracy 0.742, precision 0.438, recall 0.721, specificity 0.748, F1 score 0.545, AUC 0.787, and log loss 0.470. The model exhibits a clear tradeoff between precision and recall. The relatively high recall rate (0.721) indicates that the model successfully identifies a large population of true positives, which are good wines, while the lower precision result (0.438) shows that many wines predicted to be good by the model are actually not. This pattern arises from the relatively low classification threshold selected by Youden's method, which prioritizes sensitivity and overall balance rather than precision. In applications where false positives are costly, a higher threshold might be preferred to improve precision at the expense of recall. Conversely, if identifying as many good wines as possible is the priority, the current threshold provides a reasonable balance. Overall, these results indicate that the model maintains reasonable predictive performance on unseen data, although performance declines relative to the training set, as expected.

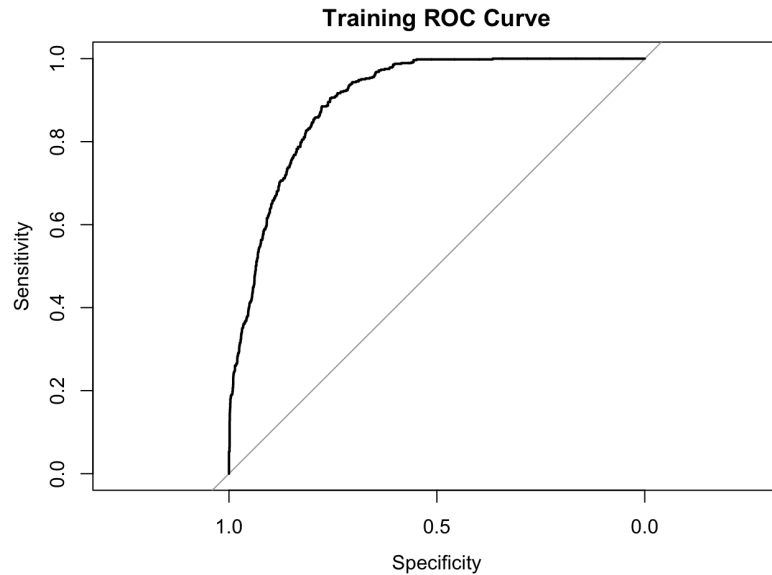


Figure 5: ROC curve on the cleaned training set. The curve is well above the diagonal, indicating good separation of good vs not-good wines.

For the final model's test-set predictions, the confusion matrix showed 864 true negatives, 227 true positives, 291 false positives, and 88 false negatives. This means the model correctly identified a substantial share of the good wines while still making a moderate number of false-positive classifications. The test ROC AUC was 0.7872, which is consistent with the earlier metric table and confirms that the final model has meaningful but not perfect discrimination on new data.

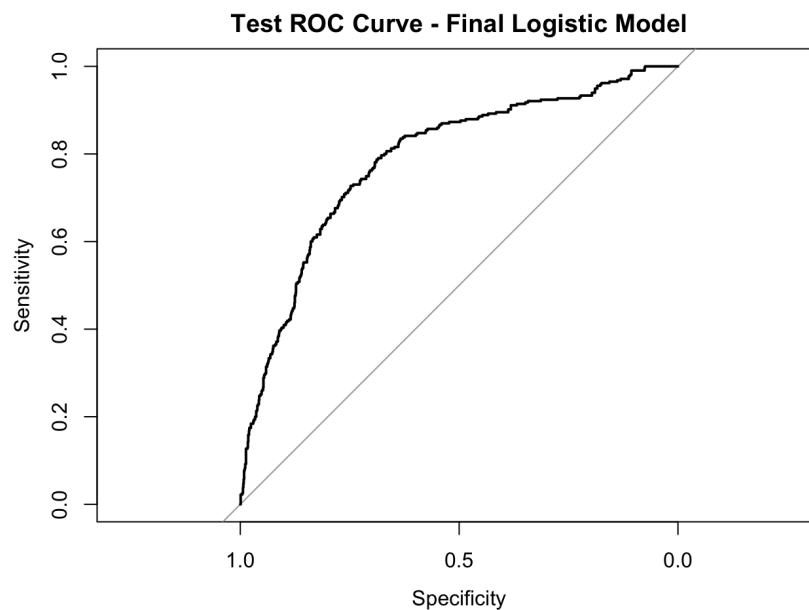


Figure 6: ROC curve on the test set. The curve remains well above the diagonal, indicating the model generalizes reasonably well to unseen data.

On the test set, the final model achieved accuracy 0.742, precision 0.438, recall 0.721, specificity 0.748, F1 score 0.545, AUC 0.787, and log loss 0.470. These results show that the model finds a substantial fraction of good wines (recall) while still making a moderate number of false positives, which is expected when the threshold is chosen to balance sensitivity and specificity. The confusion matrix results are summarized in Table 3.

Using threshold=0.5 (baseline)			Using Youden threshold=0.1485		
Actual\Predicted	0	1	Actual\Predicted	0	1
0	1067	87	0	863	291
1	201	114	1	88	227

Table 3: Confusion matrix for test-set predictions

Finally, we used the fitted final model to make a probability prediction for a wine with average values of the eight retained predictors. For this mean-profile wine, the estimated logit was -2.80 , corresponding to a predicted probability of 0.0575 that the wine would be classified as good. A 95% interval for that probability was approximately $[0.0474, 0.0697]$. Since this predicted probability is well below the chosen threshold of 0.1484781, the model classifies this average-profile wine as good = 0, meaning not good under the project's definition.

Conclusion

In this project, we used logistic regression to model whether a white wine should be classified as good using physicochemical measurements from the UCI white wine dataset. We created a binary response variable based on

whether quality was at least 7, found no missing values in the dataset, split the data into training and testing sets, and removed 208 influential observations from the training data using Cook's distance before selecting a final model. We compared a full model with stepwise AIC and BIC models and chose the BIC model because it was more parsimonious while retaining competitive fit and predictive performance.

The final BIC model retained eight predictors: chlorides, pH, volatile acidity, sulphates, free sulfur dioxide, residual sugar, density, and fixed acidity. On the training data, the model showed strong discrimination with ROC AUC 0.9018, while on the test data it achieved accuracy 0.742 and ROC AUC 0.7872. These results indicate that the model is useful for distinguishing better white wines from lower-rated ones, though its performance is not perfect. A major strength of the model is that it provides interpretable probability-based predictions for a meaningful

binary outcome. Another strength is that it reduces the full set of predictors to a simpler model without losing much predictive ability.

There are also limitations. First, converting the original quality score into a binary variable loses information about the ordering and spacing of the original ratings. Second, the Hosmer–Lemeshow test suggests that the model is not perfectly calibrated. Third, some predictors have scales that make one-unit odds-ratio interpretations difficult, so coefficient direction is often more informative than raw magnitude. Future work could compare alternative thresholds for defining a good wine, use k-fold cross-validation with $k = 5$ or 10 instead of a single train-test split, explore interaction or nonlinear terms, or fit an ordinal model that preserves more of the original quality information.

References

<https://archive.ics.uci.edu/dataset/186/wine+quality>

<https://www.geo.fu-berlin.de/en/v/soga/Basics-of-statistics/Logistic-Regression/Logistic-Regression-in->

<http://www.sthda.com/english/articles/36-classification-methods-essentials/151-logistic-regression->

Key R code

```
None

# Load packages
library(tidyverse)
library(ggpubr)
library(broom)
library(car)
library(MASS)
library(pROC)
library(pscl)
library(ResourceSelection)
theme_set(theme_pubr())

# Import data and clean column names
wine_raw <- readr::read_delim(
  "winequality-white.csv",
  delim = ";",
  show_col_types = FALSE
)
```

```

wine <- wine_raw %>%
  rename(
    fixed_acidity = `fixed acidity`,
    volatile_acidity = `volatile acidity`,
    citric_acid = `citric acid`,
    residual_sugar = `residual sugar`,
    free_sulfur_dioxide = `free sulfur dioxide`,
    total_sulfur_dioxide = `total sulfur dioxide`
  ) %>%
  mutate(
    good = ifelse(quality >= 7, 1, 0),
    good = as.integer(good)
  )

# Remove missing values
wine_clean <- wine %>% drop_na()

# Train-test split (70/30)
set.seed(456)
n <- nrow(wine_clean)
train_idx <- sample(seq_len(n), size = floor(0.70 * n))
train <- wine_clean[train_idx, ]
test <- wine_clean[-train_idx, ]

# Define formulas
predictor_names <- c(
  "fixed_acidity", "volatile_acidity", "citric_acid", "residual_sugar",
  "chlorides", "free_sulfur_dioxide", "total_sulfur_dioxide",
  "density", "pH", "sulphates", "alcohol"
)

full_formula <- as.formula(paste("good ~", paste(predictor_names, collapse = " +
")))
null_formula <- good ~ 1

# Initial full model
glm_full_train <- glm(
  full_formula,
  data = train,
  family = binomial(link = "logit")
)

# Cook's distance to remove influential points
cook_vals <- cooks.distance(glm_full_train)

```

```

cook_cutoff <- 4 / nrow(train)
influential_idx <- which(cook_vals > cook_cutoff)
n_influential <- length(influential_idx)
train_clean <- train[-influential_idx, ]

# Refit models on cleaned training data
glm_full_clean <- glm(
  full_formula,
  data = train_clean,
  family = binomial(link = "logit")
)

glm_null_clean <- glm(
  null_formula,
  data = train_clean,
  family = binomial(link = "logit")
)

# Stepwise selection (AIC and BIC)
glm_step_aic <- step(
  glm_null_clean,
  scope = list(lower = null_formula, upper = full_formula),
  direction = "both",
  trace = 0
)

glm_step_bic <- step(
  glm_null_clean,
  scope = list(lower = null_formula, upper = full_formula),
  direction = "both",
  k = log(nrow(train_clean)),
  trace = 0
)

# Select final model
glm_final <- glm_step_bic
summary(glm_final)

# VIF Check
car::vif(glm_final)

# Diagnostics
diag_df <- tibble(
  fitted = fitted(glm_final),
  deviance_resid = residuals(glm_final, type = "deviance"),

```

```

    pearson_resid = residuals(glm_final, type = "pearson")
  )

# Hosmer-Lemeshow goodness-of-fit test
hoslem_result <- hoslem.test(
  x = train_clean$good,
  y = fitted(glm_final),
  g = 10
)

# ROC curve and best threshold on training data
roc_train <- roc(train_clean$good, fitted(glm_final))
best_threshold <- as.numeric(coords(
  roc_train,
  x = "best",
  best.method = "youden",
  ret = "threshold"
))

# Evaluate on test set
test$prob_final <- predict(glm_final, newdata = test, type = "response")
test$class_final <- ifelse(test$prob_final >= best_threshold, 1, 0)

# Confusion Matrix
table(
  Actual = test$good,
  Predicted = test$class_final
)

# Test ROC
roc_test_final <- roc(test$good, test$prob_final)
auc(roc_test_final)

# Odds Ratios
coef_table <- broom::tidy(glm_final, conf.int = TRUE, exponentiate = TRUE)

# Predict for a wine with average values
final_terms <- attr(terms(glm_final), "term.labels")
new_wine <- train_clean %>%
  summarise(across(all_of(final_terms), mean))

# Calculate probabilities with 95% CI
pred_link <- predict(glm_final, newdata = new_wine, type = "link", se.fit = TRUE)
eta <- pred_link$fit[1]
se_eta <- pred_link$se.fit[1]

```

```
link_lo <- eta - 1.96 * se_eta
link_hi <- eta + 1.96 * se_eta

pred_summary <- tibble(
  logit_fit = eta,
  prob_fit = plogis(eta),
  prob_low_95 = plogis(link_lo),
  prob_high_95 = plogis(link_hi)
)

print(pred_summary)
```